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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING (24CI3201)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	ChangeLens – AI-Driven Adaptive Software Change Impact Analysis and Intelligent Dependency Management Platform	
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Abstract

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The AI-Driven Adaptive Software Change Impact Analysis and Intelligent Dependency Management Platform is an AI-integrated system designed to help software development teams understand the effects of changes made to their projects. In Agile and DevOps environments, frequent modifications to source code, requirements, APIs, and software components can unintentionally affect dependent modules and introduce errors. The proposed platform addresses this challenge by automatically analyzing software changes and identifying potentially affected components. The system integrates with GitHub to detect code changes and analyzes relationships between files, modules, APIs, and components. An AI-powered analysis layer interprets these changes and generates an impact report containing affected components, potential risks, and recommended actions. The platform can also integrate with Jira to create or update tasks related to identified changes, supporting Agile development workflows. A web-based dashboard provides developers with a clear view of changed components, dependencies, impact levels, and AI-generated recommendations. By continuously adapting its analysis to project changes, the system reduces manual dependency tracking and helps developers make informed decisions before implementing modifications. The platform demonstrates the practical application of Artificial Intelligence, Adaptive Software Engineering, Agile development, DevOps, and software maintenance, while improving change visibility, reducing potential development issues, and supporting more reliable software evolution.